Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: R.E.L.M Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6721 E Perrin RD	Manteca	San Joaquin	95337

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1949

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X257-X130-XX10-XXXX

B. OPERATORS

Machado, Edward Anthony

Operator name: Machado, Edward Anthony

Telephone no.: (209) 599-2565 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11708 E Moncure RD	Ripon	CA	95366

This operator is responsible for paying permit fees.

C. OWNERS

Teicheira, Ida

Legal owner name: Teicheira, Ida

Telephone no.: (209) 823-6658 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6721 E Perrin RD	Manteca	CA	95337

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	345	0	0	0	0	0
Maximum number	345	0	0	0	0	0
Average number	345	0	0	0	0	0
Avg live weight (lbs)	1,350	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,971.98 *tons per reporting period*

Total nitrogen from manure: 117,294.30 *lbs per reporting period*

After ammonia losses (30% loss applied): 82,106.01 *lbs per reporting period*

Total phosphorus from manure: 19,880.37 *lbs per reporting period*

Total potassium from manure: 66,750.68 *lbs per reporting period*

Total salt from manure: 162,443.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 13,928,930 *gallons*

Total nitrogen generated: 65,247.43 *lbs*

Total phosphorus generated: 8,902.65 *lbs*

Total potassium generated: 115,319.13 *lbs*

Total salt generated: 530,135.12 *lbs*

	23,972,930	*gallons applied*
+	0	*gallons exported*
-	10,044,000	*gallons imported*
=	13,928,930	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Stanislaus Water Users Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
03/25/2023	Process wastewater *WWS Imported from F&S Brasil*	3,024,000.00 *gal*	252.00	88.30	314.00		2,380
05/24/2023	Process wastewater *WWS Imported from F&S Brasil*	2,412,000.00 *gal*	459.00	50.30	588.00		4,660
07/17/2023	Process wastewater *WWS Imported from F&S Brasil*	2,304,000.00 *gal*	112.00	26.70	270.00		1,920
08/10/2023	Process wastewater *WWS Imported from F&S Brasil*	2,304,000.00 *gal*	112.00	26.70	270.00		1,920

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/06/2023	Solid commercial fertilizer *4-0-0*	36.75 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
07/03/2023	Solid commercial fertilizer *UN32*	22.61 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	17,392.99	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	19,904.93	4,267.44	30,141.74	227,688.32
Total imports for all materials	37,297.92	4,267.44	30,141.74	227,688.32

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	62	62	2	process wastewater	X257-X013-X009-XXXX X257-X013-X010-XXXX
2	26	26	2	process wastewater	X257-X013-X010-XXXX
3	29	29	2	process wastewater	X257-X013-X010-XXXX
4	32	32	2	both	X257-X013-X010-XXXX
5	20	20	2	process wastewater	X257-X013-X010-XXXX
6	8	8	2	process wastewater	X257-X013-X010-XXXX
7	70	70	2	process wastewater	X257-X013-X010-XXXX
8	47	47	2	process wastewater	X257-X013-X010-XXXX
Totals for areas that were used for application	294	294	16		
Totals for areas that were not used for application					
Land application area totals	294	294	16		

B. CROPS AND HARVESTS

1

Field name: 1

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 62 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	930.40 *ton*	As-is		58.7	4,170.00	870.00	14,000.00		7.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.01	125.15	26.11	420.18	891.22

1

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,767.00 *ton*	As-is		69.6	3,770.00	680.00	5,750.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.50	214.89	38.76	327.75	970.37

2

Field name: 2

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	395.40 *ton*	As-is		50.2	3,290.00	1,340.00	9,460.00		6.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.21	100.07	40.76	287.73	925.47

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	756.60 *ton*	As-is		69.6	3,500.00	870.00	5,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.10	203.70	50.63	308.46	1,150.03

3

Field name: 3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	440.80 *ton*	As-is		55.7	3,900.00	840.00	14,200.00		8.03

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.20	118.56	25.54	431.68	1,081.42

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	784.60 *ton*	As-is		66.4	4,570.00	1,140.00	5,240.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.06	247.28	61.69	283.54	1,236.31

4

Field name: 4

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	630.40 *ton*	As-is		67.4	3,910.00	720.00	11,100.00		7.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.70	154.05	28.37	437.34	909.38

4

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	934.40 *ton*	As-is		69.6	3,160.00	1,200.00	6,830.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.20	184.54	70.08	398.87	798.91

5

Field name: 5

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	370.00 *ton*	As-is		66.4	3,430.00	440.00	12,100.00		7.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.50	126.91	16.28	447.70	972.18

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	562.00 *ton*	As-is		69.0	3,470.00	1,010.00	5,300.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.10	195.01	56.76	297.86	888.52

6

Field name: 6

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	146.20 *ton*	As-is		66.4	3,430.00	440.00	12,100.00		7.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.28	125.37	16.08	442.26	960.36

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	232.00 *ton*	As-is		69.6	3,160.00	1,200.00	6,830.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.00	183.28	69.60	396.14	793.44

7

Field name: 7

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 70 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,059.00 *ton*	As-is		58.6	3,680.00	790.00	12,000.00		6.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.13	111.35	23.90	363.09	790.42

7

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	2,009.00 *ton*	As-is		68.4	3,480.00	1,080.00	5,550.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.70	199.75	61.99	318.57	1,233.41

8

Field name: 8

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	700.30 *ton*	As-is		58.2	4,100.00	1,170.00	11,300.00		6.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.90	122.18	34.87	336.74	758.59

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	1,334.20 *ton*	As-is		68.2	3,690.00	1,120.00	5,210.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.39	209.50	63.59	295.79	1,137.42

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/31/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.88	16.53	78.75	393.77	1,116,620.00 *gal*
Stanislaus Water Users Canal	Surface water	2.52	0.00	0.00	8.99	6,682,500.00 *gal*
Application event totals		69.40	16.53	78.75	402.76	

12/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.79	18.49	88.07	440.34	1,248,680.00 *gal*
Application event totals		74.79	18.49	88.07	440.34	

1 - 06/02/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.23	4.37	84.33	698.73	526,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.20	7,897,500.00 *gal*
Application event totals		43.23	4.37	84.33	735.94	

1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.16	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.16	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.10	2.99	94.12	329.43	499,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.30	7,492,500.00 *gal*
Application event totals		43.10	2.99	94.12	364.73	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.10	2.99	94.12	329.43	499,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.30	7,492,500.00 *gal*
Application event totals		43.10	2.99	94.12	364.73	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.31	7,495,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.31	

1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.93	2.91	91.58	320.53	486,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	34.34	7,290,000.00 *gal*
Application event totals		41.93	2.91	91.58	354.87	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.93	2.91	91.58	320.53	486,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	34.34	7,290,000.00 *gal*
Application event totals		41.93	2.91	91.58	354.87	

2 - 11/01/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.88	16.53	78.75	393.77	468,260.00 *gal*
Stanislaus Water Users Canal	Surface water	2.55	0.00	0.00	9.10	2,835,000.00 *gal*
Application event totals		69.43	16.53	78.75	402.87	

01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.13	18.87	108.41	394.65	243,000.00 *gal*
Application event totals		88.13	18.87	108.41	394.65	

2 - 06/03/2023: Corn, silage

2 - 06/03/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.61	4.40	85.08	704.93	222,750.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.53	3,341,250.00 *gal*
Application event totals		43.61	4.40	85.08	742.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.67	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.44	3.09	97.06	339.71	216,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.40	3,240,000.00 *gal*
Application event totals		44.44	3.09	97.06	376.10	

2 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.44	3.09	97.06	339.71	216,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.40	3,240,000.00 *gal*
Application event totals		44.44	3.09	97.06	376.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.53	3,341,250.00 *gal*
Application event totals		0.00	0.00	0.00	37.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.66	2.89	90.99	318.47	202,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	34.12	3,037,500.00 *gal*
Application event totals		41.66	2.89	90.99	352.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.66	2.89	90.99	318.47	202,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	34.12	3,037,500.00 *gal*
Application event totals		41.66	2.89	90.99	352.60	

3 - 11/01/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.88	16.53	78.75	393.77	522,290.00 *gal*
Stanislaus Water Users Canal	Surface water	2.77	0.00	0.00	9.91	3,442,500.00 *gal*
Application event totals		69.65	16.53	78.75	403.67	

02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.80	18.80	108.00	393.14	270,000.00 *gal*
Application event totals		87.80	18.80	108.00	393.14	

3 - 06/03/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.84	4.43	85.52	708.62	249,750.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.73	3,746,250.00 *gal*
Application event totals		43.84	4.43	85.52	746.35	

06/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

3 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.75	3,847,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.75	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.82	3.11	97.90	342.63	243,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.71	3,645,000.00 *gal*
Application event totals		44.82	3.11	97.90	379.34	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.82	3.11	97.90	342.63	243,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.71	3,645,000.00 *gal*
Application event totals		44.82	3.11	97.90	379.34	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.73	3,746,250.00 *gal*
Application event totals		0.00	0.00	0.00	37.73	

3 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.33	2.94	92.46	323.60	229,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	34.67	3,442,500.00 *gal*
Application event totals		42.33	2.94	92.46	358.27	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.33	2.94	92.46	323.60	229,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	34.67	3,442,500.00 *gal*
Application event totals		42.33	2.94	92.46	358.27	

4 - 11/02/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	248.06	108.16	306.61	4,415.51	210.00 *ton*
Application event totals		248.06	108.16	306.61	4,415.51	

10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.97	

4 - 06/03/2023: Corn, silage

4 - 06/03/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.95	4.34	83.79	694.25	270,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	4,050,000.00 *gal*
Application event totals		42.95	4.34	83.79	731.22	

06/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

07/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.81	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.81	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.88	2.98	93.65	327.76	256,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.12	3,847,500.00 *gal*
Application event totals		42.88	2.98	93.65	362.88	

4 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.88	2.98	93.65	327.76	256,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.12	3,847,500.00 *gal*
Application event totals		42.88	2.98	93.65	362.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.89	4,151,250.00 *gal*
Application event totals		0.00	0.00	0.00	37.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.88	2.98	93.65	327.76	256,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.12	3,847,500.00 *gal*
Application event totals		42.88	2.98	93.65	362.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.97	

5 - 11/02/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.88	16.53	78.75	393.77	360,200.00 *gal*
Stanislaus Water Users Canal	Surface water	2.60	0.00	0.00	9.29	2,227,500.00 *gal*
Application event totals		69.48	16.53	78.75	403.06	

12/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.79	18.49	88.07	440.34	402,800.00 *gal*
Application event totals		74.79	18.49	88.07	440.34	

5 - 06/04/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.95	4.34	83.79	694.25	168,750.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	2,531,250.00 *gal*
Application event totals		42.95	4.34	83.79	731.22	

06/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.44	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.44	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.33	3.01	94.63	331.21	162,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.49	2,430,000.00 *gal*
Application event totals		43.33	3.01	94.63	366.70	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.33	3.01	94.63	331.21	162,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.49	2,430,000.00 *gal*
Application event totals		43.33	3.01	94.63	366.70	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.44	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.44	

5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.33	3.01	94.63	331.21	162,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	35.49	2,430,000.00 *gal*
Application event totals		43.33	3.01	94.63	366.70	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	2,531,250.00 *gal*
Application event totals		0.00	0.00	0.00	36.97	

6 - 11/02/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.88	16.53	78.75	393.77	144,080.00 *gal*
Stanislaus Water Users Canal	Surface water	2.66	0.00	0.00	9.51	911,250.00 *gal*
Application event totals		69.54	16.53	78.75	403.27	

02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.48	20.45	117.45	427.54	81,000.00 *gal*
Application event totals		95.48	20.45	117.45	427.54	

6 - 06/04/2023: Corn, silage

6 - 06/04/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.95	4.34	83.79	694.25	67,500.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	1,012,500.00 *gal*
Application event totals		42.95	4.34	83.79	731.22	

06/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

07/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	3.25	0.00	0.00	11.62	1,113,750.00 *gal*
Application event totals		3.25	0.00	0.00	11.62	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.62	2.82	88.72	310.51	60,750.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	33.27	911,250.00 *gal*
Application event totals		40.62	2.82	88.72	343.78	

6 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.62	2.82	88.72	310.51	60,750.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	33.27	911,250.00 *gal*
Application event totals		40.62	2.82	88.72	343.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	36.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.62	2.82	88.72	310.51	60,750.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	33.27	911,250.00 *gal*
Application event totals		40.62	2.82	88.72	343.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.97	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	36.97	

7 - 11/04/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.62	9.42	122.91	655.77	1,440,000.00 *gal*
Stanislaus Water Users Canal	Surface water	2.70	0.00	0.00	9.66	8,100,000.00 *gal*
Application event totals		75.32	9.42	122.91	665.43	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.91	18.19	64.68	490.29	1,728,000.00 *gal*
Application event totals		51.91	18.19	64.68	490.29	

7 - 06/05/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.80	8.63	100.94	799.98	1,440,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	33.80	8,100,000.00 *gal*
Application event totals		78.80	8.63	100.94	833.77	

06/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

7 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	70.40	0.00	0.00	0.00	
Application event totals		70.40	0.00	0.00	0.00	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	38.02	9,112,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.02	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.27	4.35	44.03	313.12	1,368,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	32.11	7,695,000.00 *gal*
Application event totals		18.27	4.35	44.03	345.23	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.18	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.18	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.27	4.35	44.03	313.12	1,368,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	32.11	7,695,000.00 *gal*
Application event totals		18.27	4.35	44.03	345.23	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.18	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.18	

7 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.18	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.18	

8 - 11/03/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.30	9.12	118.99	634.84	936,000.00 *gal*
Stanislaus Water Users Canal	Surface water	2.62	0.00	0.00	9.35	5,265,000.00 *gal*
Application event totals		72.92	9.12	118.99	644.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.99	20.32	72.25	547.66	1,296,000.00 *gal*
Application event totals		57.99	20.32	72.25	547.66	

8 - 06/06/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

8 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.22	8.68	101.48	804.23	972,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	33.98	5,467,500.00 *gal*
Application event totals		79.22	8.68	101.48	838.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	70.40	0.00	0.00	0.00	
Application event totals		70.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.75	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.61	4.44	44.87	319.08	936,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	32.72	5,265,000.00 *gal*
Application event totals		18.61	4.44	44.87	351.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	32.72	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.72	

8 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.61	4.44	44.87	319.08	936,000.00 *gal*
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	32.72	5,265,000.00 *gal*
Application event totals		18.61	4.44	44.87	351.80	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	37.75	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.75	

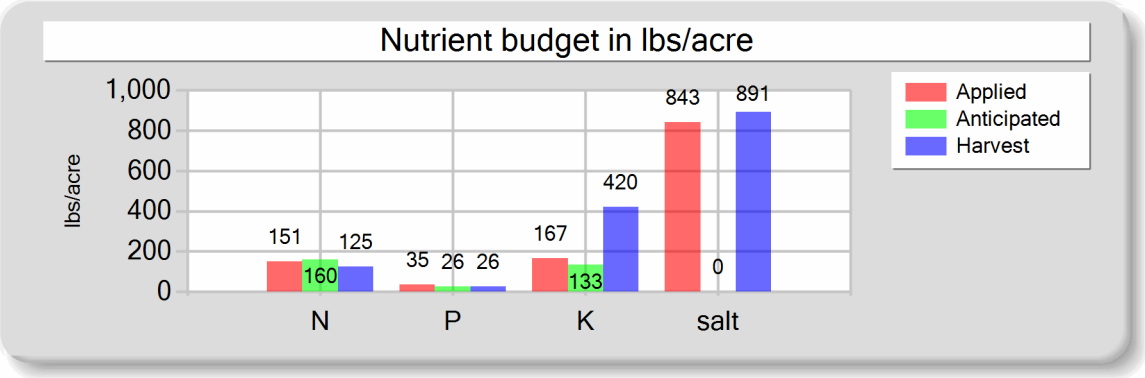
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	36.49	5,872,500.00 *gal*
Application event totals		0.00	0.00	0.00	36.49	

B. NUTRIENT BUDGET

1 - 10/31/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.67	35.02	166.82	834.11
Fresh water	2.52	0.00	0.00	8.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.19	35.02	166.82	843.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.15	26.11	420.18	891.22
Nutrient balance	26.04	8.91	-253.36	-48.12
Applied to removed ratio	1.21	1.34	0.40	0.95

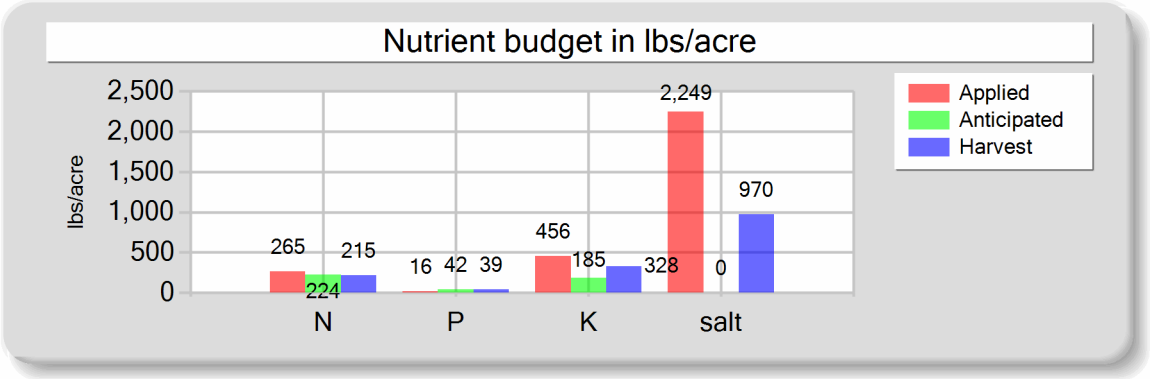
Fresh water applied	
6,682,500.00	*gallons*
246.09	*acre-inches*
3.97	*inches/acre*

Process wastewater applied	
2,365,300.00	*gallons*
87.11	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/02/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.28	16.17	455.74	1,998.65
Fresh water	0.00	0.00	0.00	249.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.48	16.17	455.74	2,248.60
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	214.89	38.76	327.75	970.37
Nutrient balance	50.59	-22.59	127.99	1,278.23
Applied to removed ratio	1.24	0.42	1.39	2.32

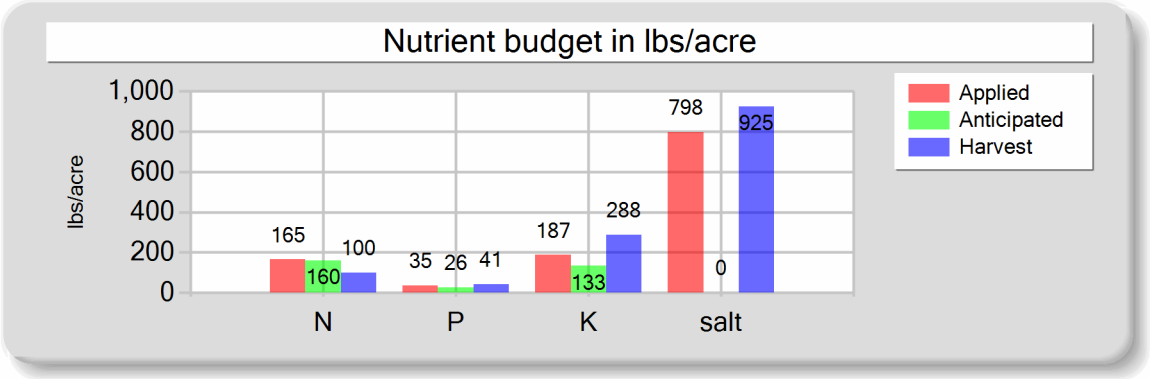
Fresh water applied	
53,057,500.00	*gallons*
1,953.93	*acre-inches*
31.51	*inches/acre*

Process wastewater applied	
2,497,500.00	*gallons*
91.97	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/01/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	155.01	35.41	187.16	788.42
Fresh water	2.55	0.00	0.00	9.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	164.56	35.41	187.16	797.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	100.07	40.76	287.73	925.47
Nutrient balance	64.49	-5.35	-100.56	-127.96
Applied to removed ratio	1.64	0.87	0.65	0.86

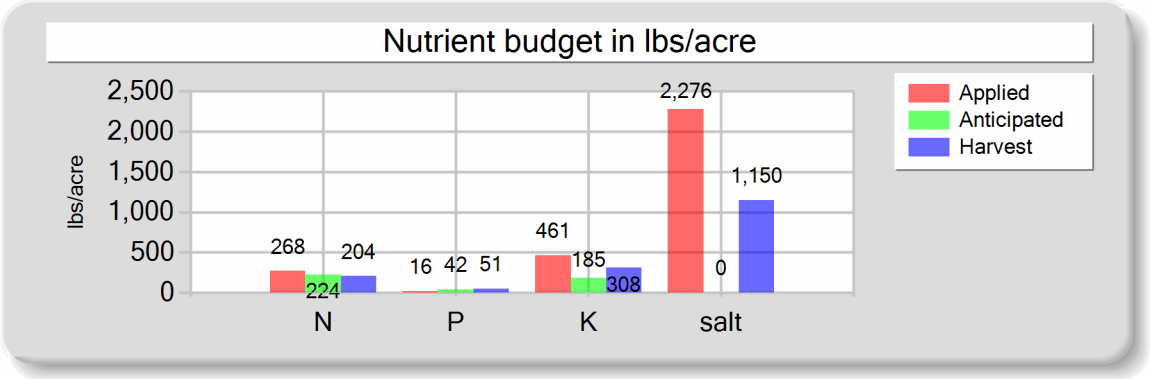
Fresh water applied	
2,835,000.00	*gallons*
104.40	*acre-inches*
4.02	*inches/acre*

Process wastewater applied	
711,260.00	*gallons*
26.19	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/03/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.81	16.36	461.18	2,021.29
Fresh water	0.00	0.00	0.00	254.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.01	16.36	461.18	2,276.07
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	203.70	50.63	308.46	1,150.03
Nutrient balance	64.31	-34.28	152.72	1,126.04
Applied to removed ratio	1.32	0.32	1.50	1.98

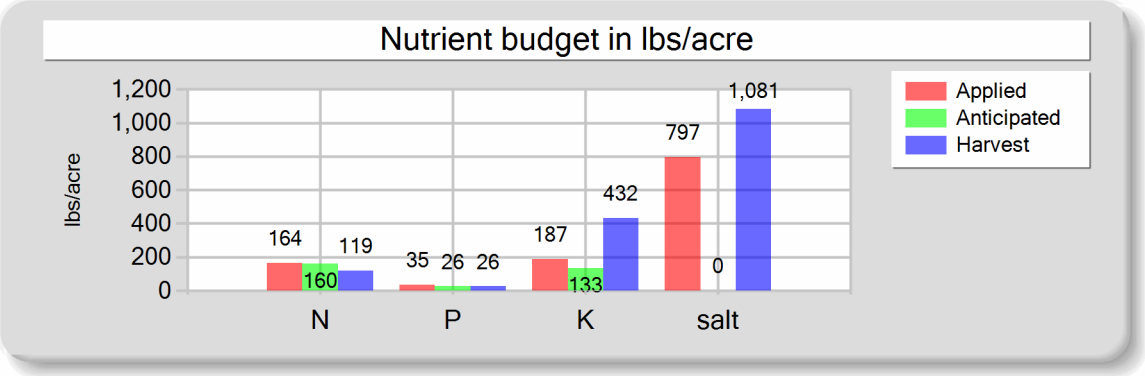
Fresh water applied	
22,680,000.00	*gallons*
835.23	*acre-inches*
32.12	*inches/acre*

Process wastewater applied	
1,059,750.00	*gallons*
39.03	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/01/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.68	35.33	186.75	786.90
Fresh water	2.77	0.00	0.00	9.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	164.45	35.33	186.75	796.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	118.56	25.54	431.68	1,081.42
Nutrient balance	45.89	9.80	-244.93	-284.61
Applied to removed ratio	1.39	1.38	0.43	0.74

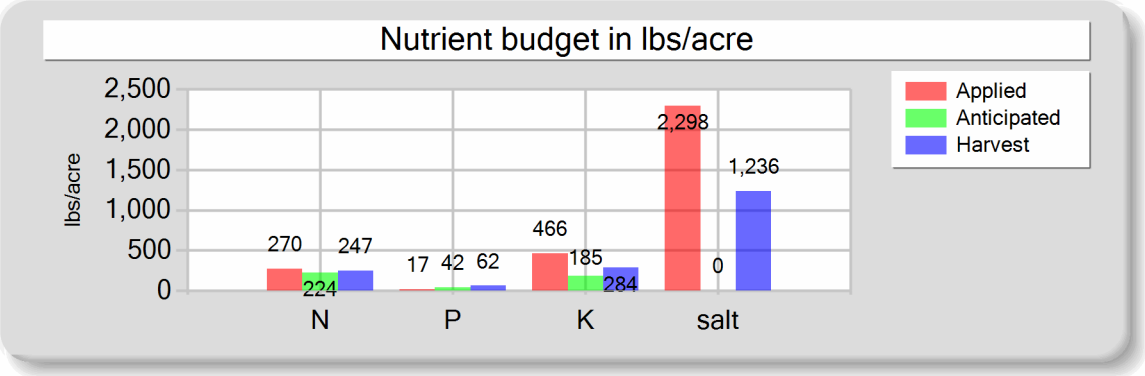
Fresh water applied	
3,442,500.00	*gallons*
126.78	*acre-inches*
4.37	*inches/acre*

Process wastewater applied	
792,290.00	*gallons*
29.18	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/03/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.15	16.53	466.23	2,041.08
Fresh water	0.00	0.00	0.00	256.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.35	16.53	466.23	2,298.06
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	247.28	61.69	283.54	1,236.31
Nutrient balance	23.06	-45.16	182.69	1,061.74
Applied to removed ratio	1.09	0.27	1.64	1.86

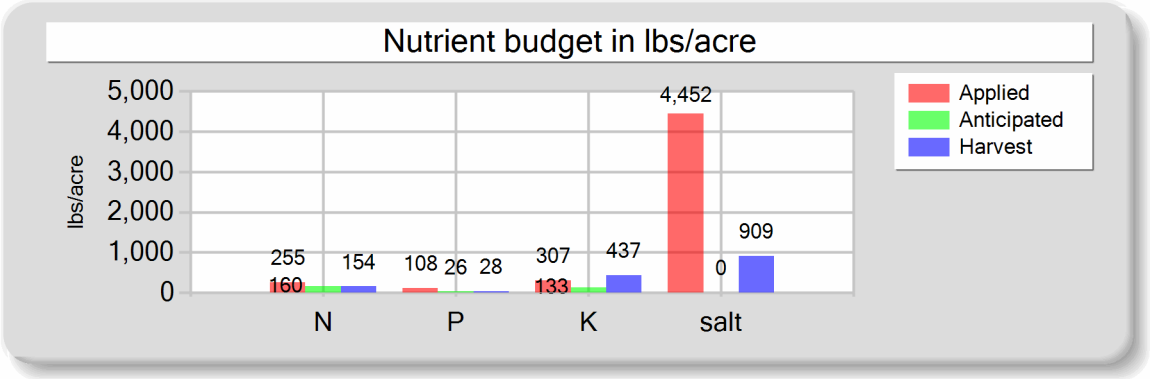
Fresh water applied	
25,515,000.00	*gallons*
939.63	*acre-inches*
32.40	*inches/acre*

Process wastewater applied	
1,194,750.00	*gallons*
44.00	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/02/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	248.06	108.16	306.61	4,415.51
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	36.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.06	108.16	306.61	4,452.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.05	28.37	437.34	909.38
Nutrient balance	101.01	79.79	-130.73	3,543.09
Applied to removed ratio	1.66	3.81	0.70	4.90

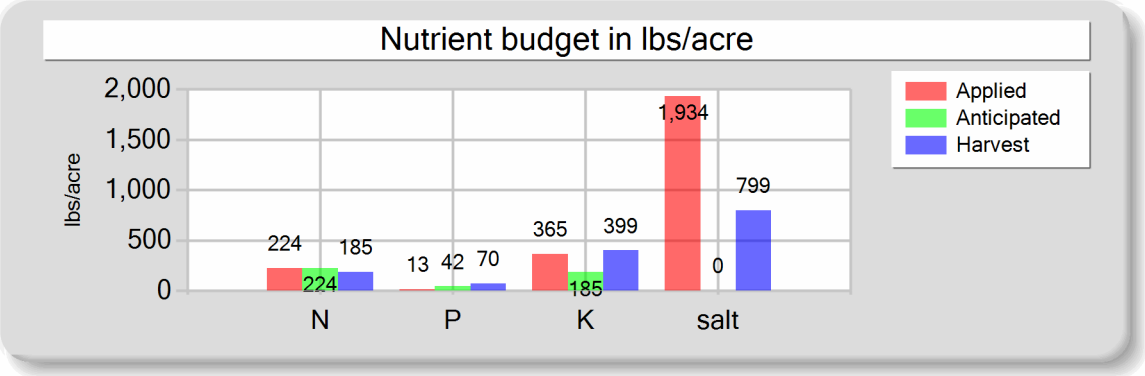
Fresh water applied	
4,050,000.00	*gallons*
149.15	*acre-inches*
4.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/03/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.58	13.27	364.73	1,677.54
Fresh water	0.00	0.00	0.00	255.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.78	13.27	364.73	1,933.53
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	184.54	70.08	398.87	798.91
Nutrient balance	39.24	-56.81	-34.14	1,134.62
Applied to removed ratio	1.21	0.19	0.91	2.42

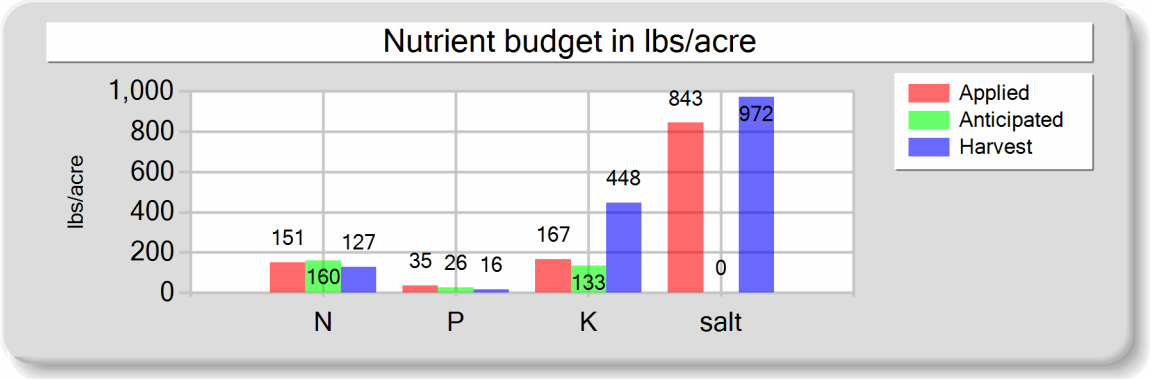
Fresh water applied	
28,046,250.00	*gallons*
1,032.85	*acre-inches*
32.28	*inches/acre*

Process wastewater applied	
1,039,500.00	*gallons*
38.28	*acre-inches*
1.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/02/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.67	35.02	166.82	834.11
Fresh water	2.60	0.00	0.00	9.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.27	35.02	166.82	843.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	126.91	16.28	447.70	972.18
Nutrient balance	24.36	18.74	-280.88	-128.78
Applied to removed ratio	1.19	2.15	0.37	0.87

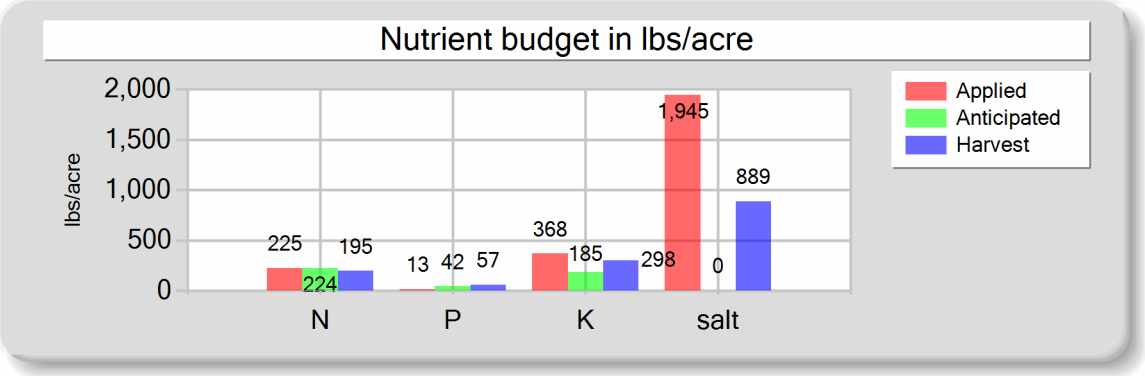
Fresh water applied	
2,227,500.00	*gallons*
82.03	*acre-inches*
4.10	*inches/acre*

Process wastewater applied	
763,000.00	*gallons*
28.10	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/04/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	172.93	13.36	367.69	1,687.89
Fresh water	0.00	0.00	0.00	257.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.13	13.36	367.69	1,945.17
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	195.01	56.76	297.86	888.52
Nutrient balance	30.12	-43.40	69.83	1,056.65
Applied to removed ratio	1.15	0.24	1.23	2.19

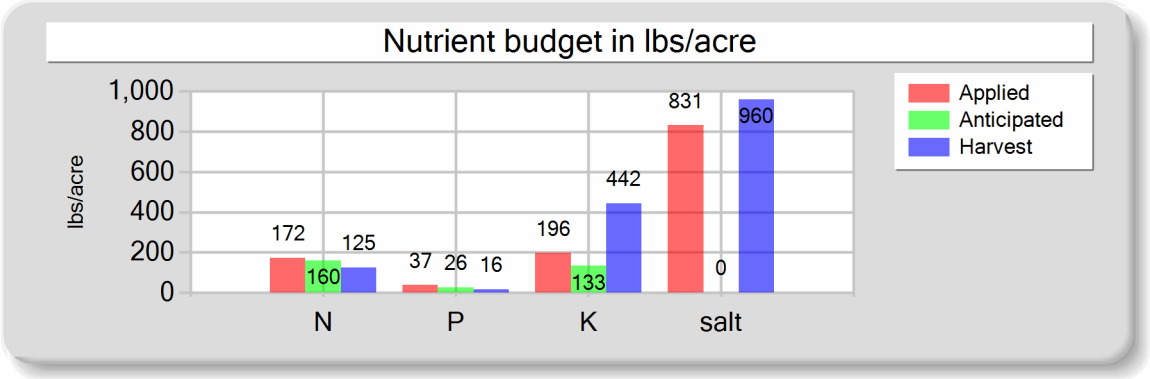
Fresh water applied	
17,617,500.00	*gallons*
648.79	*acre-inches*
32.44	*inches/acre*

Process wastewater applied	
654,750.00	*gallons*
24.11	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/02/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	162.36	36.98	196.20	821.30
Fresh water	2.66	0.00	0.00	9.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.02	36.98	196.20	830.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.37	16.08	442.26	960.36
Nutrient balance	46.65	20.90	-246.06	-129.55
Applied to removed ratio	1.37	2.30	0.44	0.87

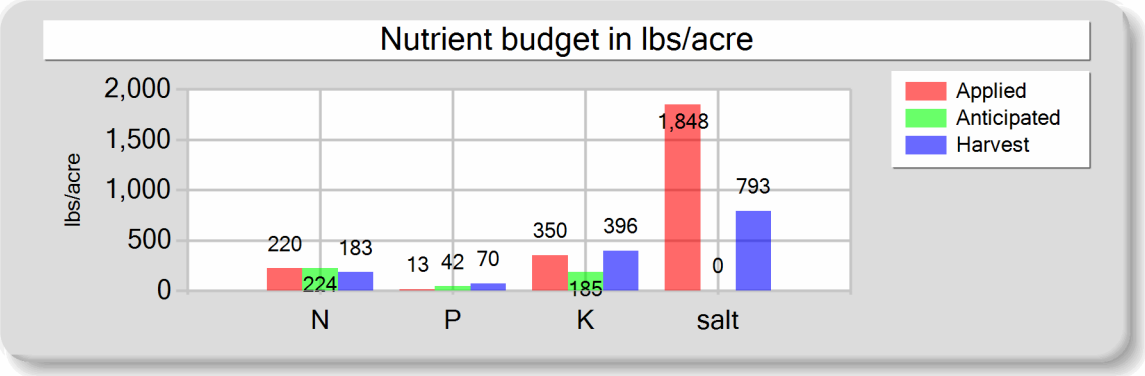
Fresh water applied	
911,250.00	*gallons*
33.56	*acre-inches*
4.19	*inches/acre*

Process wastewater applied	
225,080.00	*gallons*
8.29	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/04/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.81	12.80	349.94	1,625.79
Fresh water	3.25	0.00	0.00	222.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.26	12.80	349.94	1,848.11
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	183.28	69.60	396.14	793.44
Nutrient balance	36.98	-56.80	-46.20	1,054.67
Applied to removed ratio	1.20	0.18	0.88	2.33

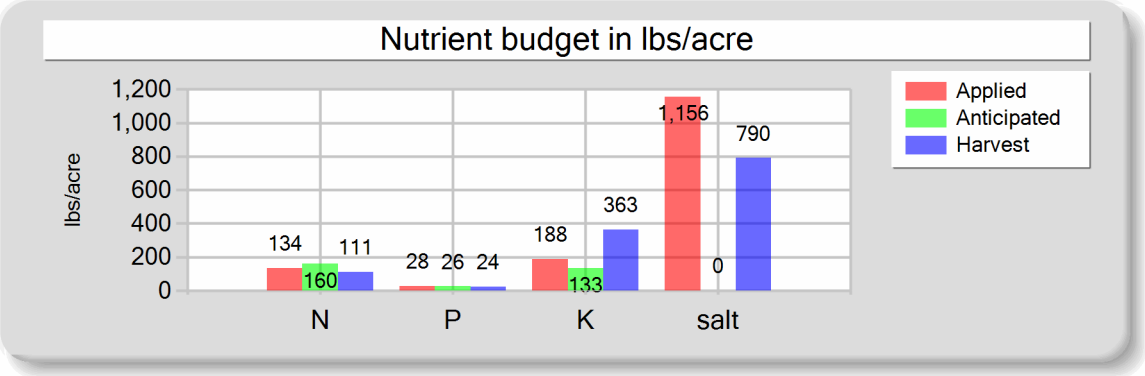
Fresh water applied	
6,885,000.00	*gallons*
253.55	*acre-inches*
31.69	*inches/acre*

Process wastewater applied	
249,750.00	*gallons*
9.20	*acre-inches*
1.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/04/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	124.53	27.61	187.60	1,146.06
Fresh water	2.70	0.00	0.00	9.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	134.23	27.61	187.60	1,155.72
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	111.35	23.90	363.09	790.42
Nutrient balance	22.89	3.71	-175.49	365.30
Applied to removed ratio	1.21	1.16	0.52	1.46

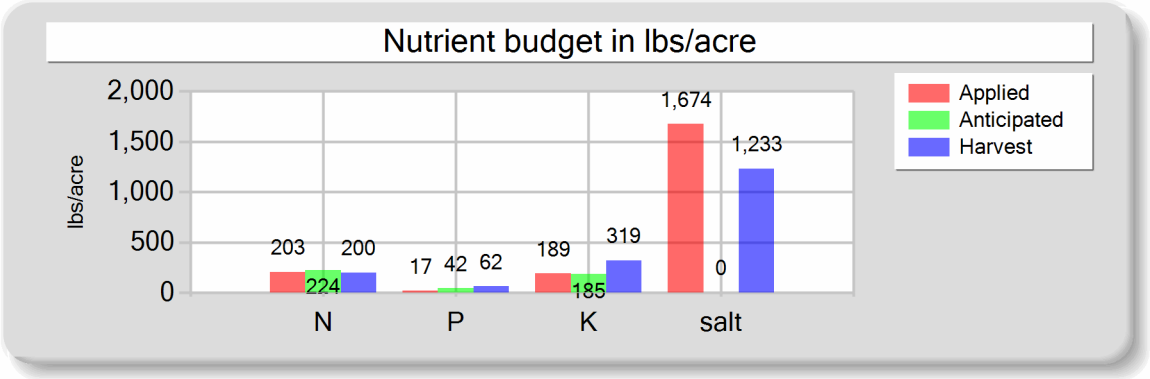
Fresh water applied	
8,100,000.00	*gallons*
298.30	*acre-inches*
4.26	*inches/acre*

Process wastewater applied	
3,168,000.00	*gallons*
116.67	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/05/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	115.33	17.34	189.01	1,426.22
Fresh water	0.00	0.00	0.00	247.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.73	17.34	189.01	1,673.79
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	199.75	61.99	318.57	1,233.41
Nutrient balance	2.97	-44.65	-129.56	440.38
Applied to removed ratio	1.01	0.28	0.59	1.36

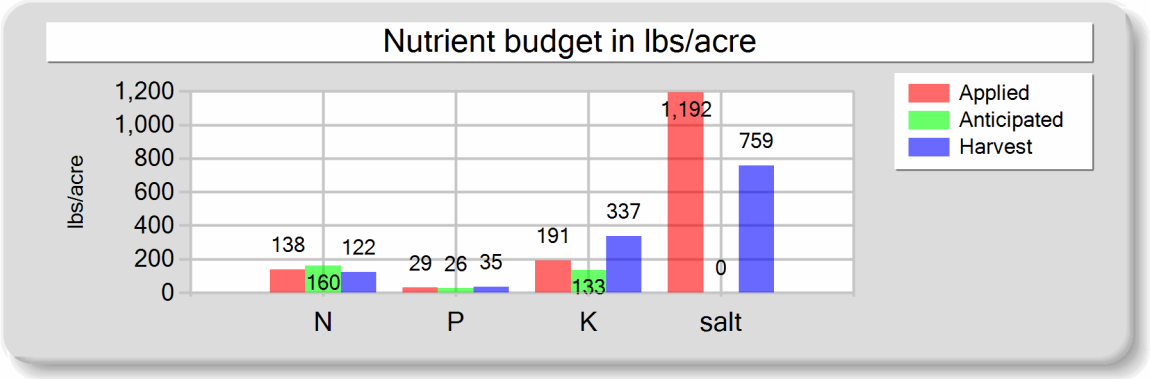
Fresh water applied	
59,332,500.00	*gallons*
2,185.01	*acre-inches*
31.21	*inches/acre*

Process wastewater applied	
4,176,000.00	*gallons*
153.79	*acre-inches*
2.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/03/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	128.29	29.44	191.25	1,182.50
Fresh water	2.62	0.00	0.00	9.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	137.90	29.44	191.25	1,191.85
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	122.18	34.87	336.74	758.59
Nutrient balance	15.72	-5.42	-145.49	433.26
Applied to removed ratio	1.13	0.84	0.57	1.57

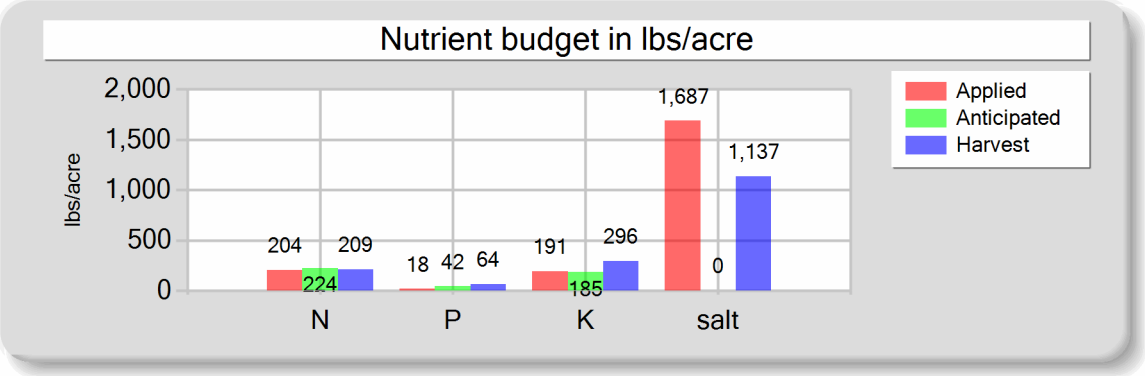
Fresh water applied	
5,265,000.00	*gallons*
193.89	*acre-inches*
4.13	*inches/acre*

Process wastewater applied	
2,232,000.00	*gallons*
82.20	*acre-inches*
1.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/06/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	116.44	17.56	191.22	1,442.40
Fresh water	0.00	0.00	0.00	244.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.84	17.56	191.22	1,686.53
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	209.50	63.59	295.79	1,137.42
Nutrient balance	-5.66	-46.03	-104.57	549.11
Applied to removed ratio	0.97	0.28	0.65	1.48

Fresh water applied	
39,285,000.00	*gallons*
1,446.73	*acre-inches*
30.78	*inches/acre*

Process wastewater applied	
2,844,000.00	*gallons*
104.73	*acre-inches*
2.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	10,900.00	30,900.00	0.00	0.00	0.00	0.00	0.00		44.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	11,500.00	36,500.00	0.00	0.00	0.00	0.00	0.00		58.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/31/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,200.00	14,100.00	23,100.00	0.00	0.00	0.00	0.00	0.00		67.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	445.00	185.00	0.00	0.00	110.00	524.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,650.00	2,620
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4 Imported from F&S Brasil

Sample and source description: WWQ4 Imported from F&S Brasil

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	423.00	238.00	0.00	0.00	54.90	716.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,050.00	3,820
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1 Import from F&S Brasil

Sample and source description: WWQ1 Import from F&S Brasil

Sample date: 01/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	252.00	129.00	0.00	0.00	88.30	314.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3,060.00	2,380
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,130.00	403.00	0.00	0.00	242.00	1,390.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,910.00	5,060
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	610.00	546.00	0.00	0.00	61.60	1,190.00	141.00	101.00	276.00	4,460.00	0.00	276.00	411.00	8,680.00	9,860
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2 Import from F&S Brasil

Sample and source description: WWQ2 Import from F&S Brasil

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	459.00	316.00	0.00	0.00	50.30	588.00	160.00	103.00	179.00	2,860.00	0.00	303.00	307.00	5,360.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	641.00	510.00	0.00	0.00	44.50	1,400.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9,210.00	4,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	100.00	10

WWQ3 Imported from F&S Brasil

Sample and source description: WWQ3 Imported from F&S Brasil

Sample date: 08/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	112.00	53.20	0.00	0.00	26.70	270.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,440.00	1,920
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	526.00	286.00	0.00	0.00	49.70	669.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,420.00	4,720
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

Stanislaus Water Users Canal

Canal

Sample description: Canal

Sample date: 08/03/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	10
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

Canal

Sample description: Canal

Sample date: 07/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.00	35
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

1

1

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			43.00				
DL			4.00				

2

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			45.00				
DL			4.00				

3

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			22.00				
DL			4.00				

4

4

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			80.00				
DL			4.00				

5

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			4.00				

6

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

7

7

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			65.00				
DL			4.00				

8

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			54.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	870.00	14,000.00		7.19
DL	100.00	100.00	100.00		0.01

1 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	1,220.00	12,700.00		5.99
DL	100.00	100.00	100.00		0.01

1 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,470.00	600.00	5,620.00		5.10
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,770.00	680.00	5,750.00		5.60
DL	100.00	100.00	100.00		0.01

2 - 11/01/2022: Oats, silage-soft dough

2 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 50.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,290.00	1,340.00	9,460.00		6.11
DL	100.00	100.00	100.00		0.01

2 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	870.00	5,300.00		6.50
DL	100.00	100.00	100.00		0.01

3 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	840.00	14,200.00		8.03
DL	100.00	100.00	100.00		0.01

3 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,570.00	1,140.00	5,240.00		6.80
DL	100.00	100.00	100.00		0.01

4 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,910.00	720.00	11,100.00		7.08
DL	100.00	100.00	100.00		0.01

4 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,160.00	1,200.00	6,830.00		4.50
DL	100.00	100.00	100.00		0.01

5 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,430.00	440.00	12,100.00		7.82
DL	100.00	100.00	100.00		0.01

5 - 06/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,470.00	1,010.00	5,300.00		5.10
DL	100.00	100.00	100.00		0.01

6 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,430.00	440.00	12,100.00		7.82
DL	100.00	100.00	100.00		0.01

6 - 06/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,160.00	1,200.00	6,830.00		4.50
DL	100.00	100.00	100.00		0.01

7 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,680.00	790.00	12,000.00		6.31
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,280.00	1,010.00	11,300.00		7.20
DL	100.00	100.00	100.00		0.01

7 - 06/05/2023: Corn, silage

7 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,480.00	1,080.00	5,550.00		6.80
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,250.00	890.00	4,650.00		5.10
DL	100.00	100.00	100.00		0.01

8 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	1,440.00	12,500.00		6.61
DL	100.00	100.00	100.00		0.01

8 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,170.00	11,300.00		6.09
DL	100.00	100.00	100.00		0.01

8 - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,750.00	1,040.00	6,140.00		5.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,690.00	1,120.00	5,210.00		6.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

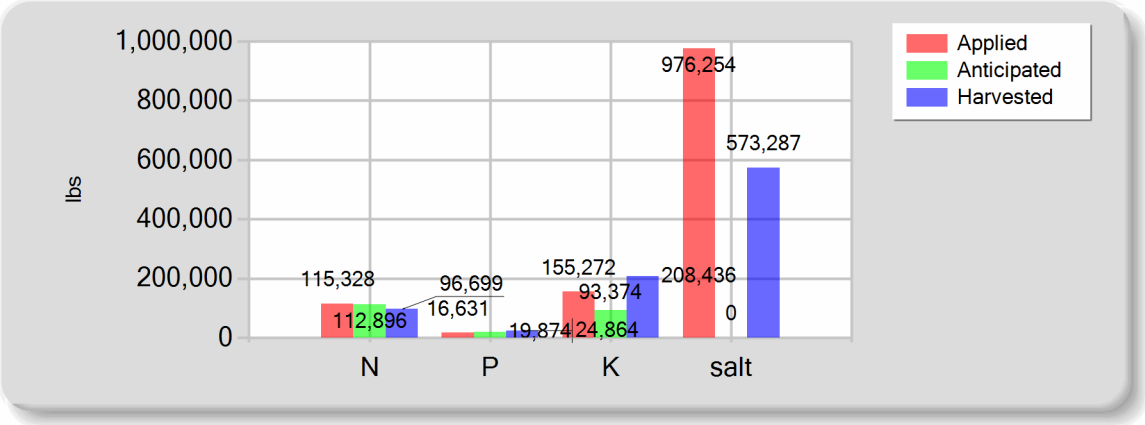
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

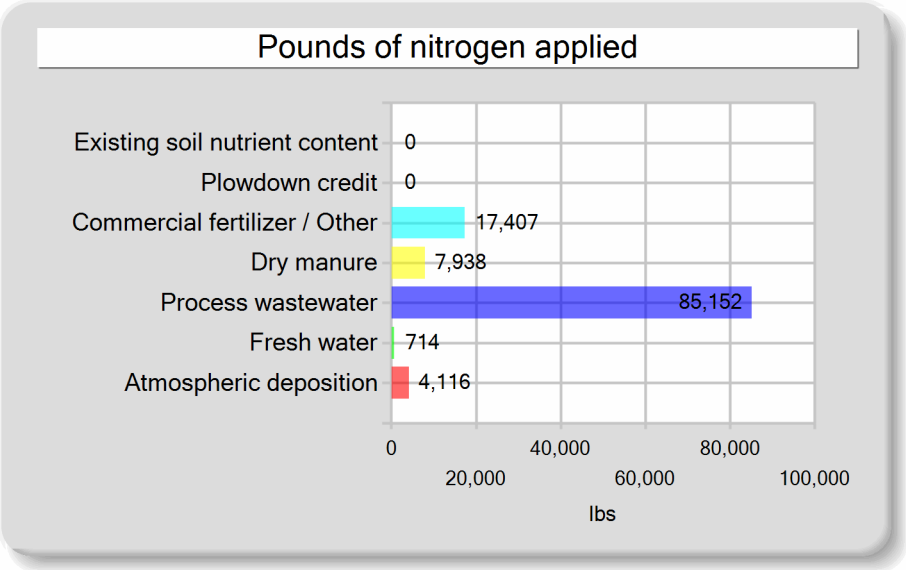
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

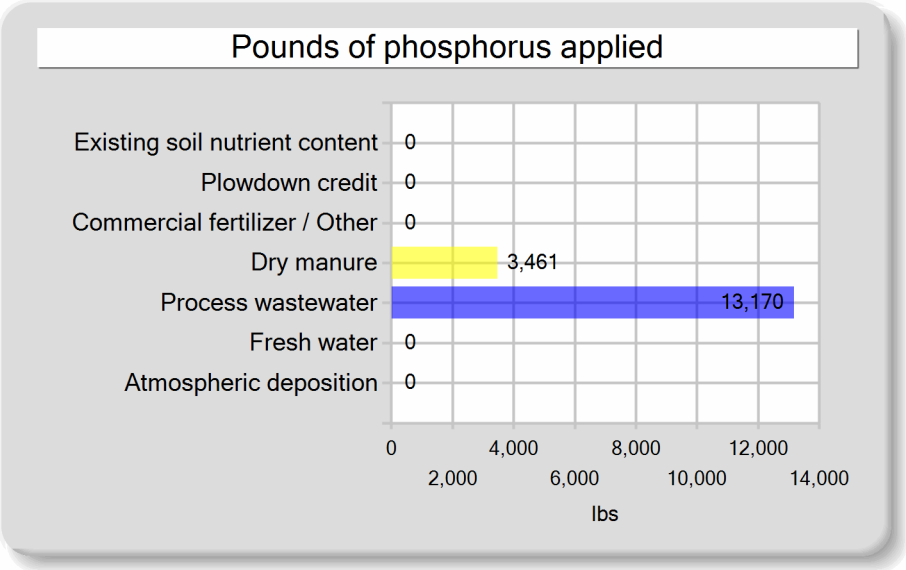
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	17,407.20	0.00	0.00	0.00
Dry manure	7,938.00	3,460.97	9,811.37	141,296.40
Process wastewater	85,152.36	13,170.09	145,460.87	757,823.44
Fresh water	714.47	0.00	0.00	77,134.50
Atmospheric deposition	4,116.00	0.00	0.00	0.00
Total nutrients applied	115,328.03	16,631.05	155,272.24	976,254.35
Anticipated crop nutrient removal	112,896.00	19,874.40	93,374.40	0.00
Actual crop nutrient removal	96,698.66	24,864.21	208,436.14	573,286.76
Nutrient balance	18,629.37	-8,233.15	-53,163.91	402,967.59
Applied to removed ratio	1.19	0.67	0.74	1.70

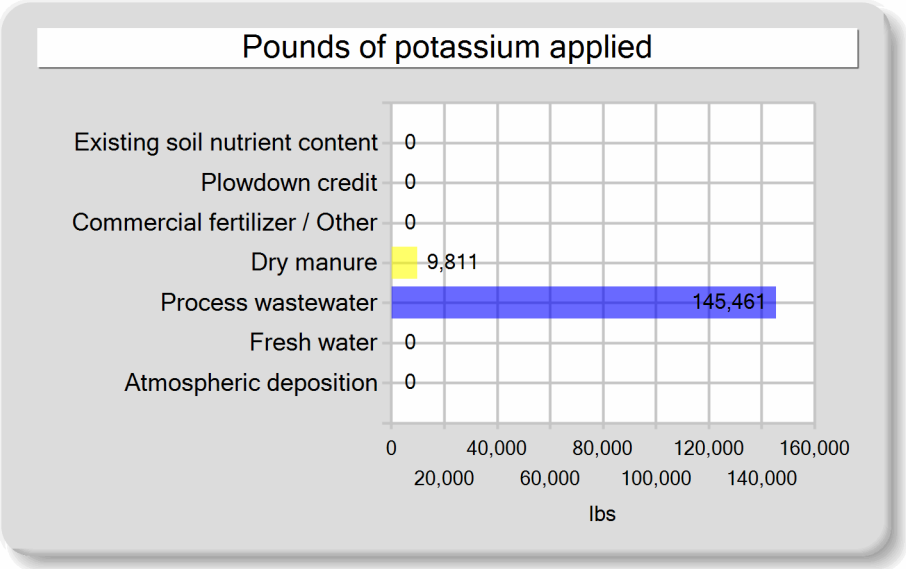
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

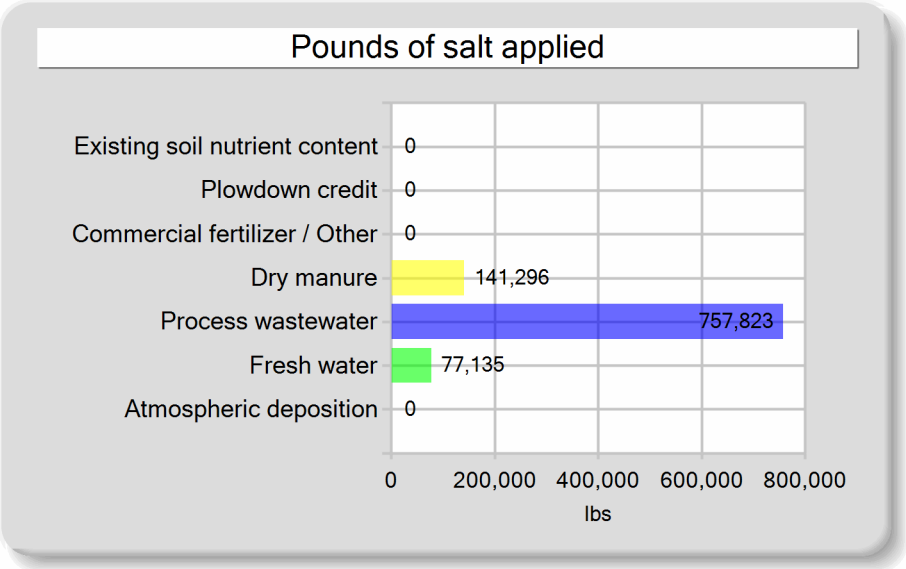


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 2 Oats (1.64) and Field 4 Oats (1.67) had high nitrogen field ratios as crops failed to meet anticipated nitrogen removal of 160 pounds/acre. The subject fields removed 100.1 and 154.1 pounds of nitrogen respectively. With regard to Field 2, yield was slightly below expected levels however a low nitrogen value from the plant tissue lab analysis resulted in 6.58 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen per harvested ton which ultimately resulted in the high nitrogen field ratio. Nitrogen application to Field 2 oats was not considered excessive but in line with that recommended in the dairy's Nutrient Management Plan. With regard to Field 4 Oats, yield was above average however a low nitrogen lab value from the plant tissue analysis resulted in 7.82 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removal per harvested ton which ultimately was one contributing factor for the high nitrogen field ratio. The other contributing factor for the high nitrogen field ratio on Field 4 Oats was slight over-application of nitrogen in the form of solid manure. RELM Dairy will need to adjust solid manure application levels in the future in order to remain under the 1.4 field ratio limit.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ida M Teicheira	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ida Teicheira	Edward Anthony Machado
PRINT OR TYPE NAME	PRINT OR TYPE NAME
2/9/24	2/9/2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
RELM
11708 E. Moncure Rd.
Ripon, CA 95366

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 7/27/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 7/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 26.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well	T3208033A	Electrical Conductivity (EC)	EPA 120.1 and	0.859	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.93	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	550	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.8			
Electrical Conductivity (EC)	98.8			0.000
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)	0.538

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: RELM
Mailing address:
(not facility address) RIPON CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	7/27/23	9:40	DOMESTIC WELL	DOMESTIC	26.0°		X				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	7-27-23 3:00			T3208033

Required to be submitted to any agency? (YES / NO)